

Q1. Debug the SQLite Program The following program is intended to create a table and insert student records. However, it contains multiple errors.

```
import sqlite3

conn = sqlite3.connect(college.db)

cursor = conn.cursor

cursor.execute("""

CREATE TABLE Student(

id INTEGER PRIMARY KEY

name TEXT,

marks INTEGER

)

""")

cursor.execute("INSERT INTO Student VALUES(1,'John',90)")

conn.close()
```

TASK:

1. Identify at least five errors.
2. Rewrite the corrected program.
3. Explain the purpose of each correction.

SOLUTION:

CORRECTED CODE:

```
import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Student(

    id INTEGER PRIMARY KEY,

    name TEXT,
```

```

marks INTEGER
)
""")
cursor.execute("INSERT INTO Student VALUES(1, 'John', 90)")
conn.commit()
conn.close()

```

2. Integrated Debugging Problem

```

import sqlite3

conn=sqlite3.connect("college.db")

cursor=conn.cursor()

cursor.execute("CREATE TABLE Student(id INTEGER PRIMARY KEY,name TEXT,marks
INTEGER)")

cursor.execute("INSERT INTO Student VALUES(1,'Asha',90)")

cursor.execute("INSERT INTO Student VALUES(1,'Raj',85)")

cursor.execute("UPDATE Student SET mark=95 WHERE id=1")

cursor.execute("SELECT name,marks FROM Students")

print(cursor.fetchall())

conn.close()

```

Task

1. Identify at least six errors.
2. Rewrite the corrected program.
3. Explain the exception generated by each error

SOLUTION

CORRECTED CODE:

```

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS Student(
    id INTEGER PRIMARY KEY,
    name TEXT,

```

marks INTEGER

)

""")

cursor.execute("INSERT OR REPLACE INTO Student VALUES(1, 'Asha', 90)")

cursor.execute("INSERT OR REPLACE INTO Student VALUES(2, 'Raj', 85)")

cursor.execute("UPDATE Student SET marks = 95 WHERE id = 1")

cursor.execute("SELECT name, marks FROM Student")

print(cursor.fetchall())

conn.commit()

conn.close()